

MS&E 228: Applied Causal Inference Powered by ML and AI

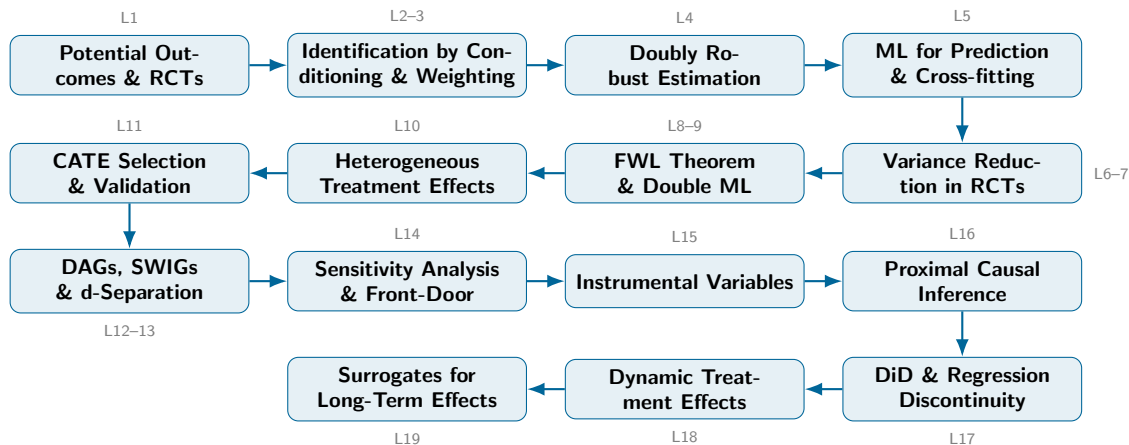
Course Summary, Q&A and Farewell

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The Journey We Took Together



Module I: Foundations of Causal Inference (Lectures 1–4)

- ▶ **Potential outcomes:** causal effects as contrasts $Y(1) - Y(0)$
- ▶ **Key estimands:** ATE, ATT, CATE
- ▶ **Identification strategies:**
 - ▶ **Conditioning:** g-formula under conditional ignorability
 - ▶ **Reweighting:** IPW / Horvitz-Thompson estimator
- ▶ **Bad controls & collider bias:** conditioning on post-treatment variables can introduce bias
- ▶ **Doubly robust estimation:** combines outcome regression and propensity weighting; consistent if *either* model is correct

Core Formula: DR Estimator

$$\hat{\tau}_{DR} = \frac{1}{n} \sum_{i=1}^n \hat{\psi}_i$$

where the pseudo-outcome is:

$$\begin{aligned} \hat{\psi}_i = & \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) \\ & + \frac{D_i(Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)} \\ & - \frac{(1 - D_i)(Y_i - \hat{\mu}_0(X_i))}{1 - \hat{e}(X_i)} \end{aligned}$$

Module II: ML-Powered Estimation (Lectures 5–9)

- ▶ **Prediction as a building block:** CEF estimation via modern ML (forests, boosting, neural nets)
- ▶ **Cross-fitting:** avoids overfitting bias when plugging ML into causal estimators
- ▶ **Variance of DR:** semiparametric efficiency bound; regression adjustment reduces variance in RCTs
- ▶ **FWL Theorem:** OLS coefficient = regression of residualized Y on residualized D
- ▶ **Double ML:** residualize with ML, then regress; $\sqrt{n}$ -consistent under nuisance rate conditions

DML Algorithm (Sketch)

1. Split data into K folds
2. For each fold k :
 - ▶ Train $\hat{m}_{-k}(X)$ for $\mathbb{E}[Y|X]$
 - ▶ Train $\hat{\ell}_{-k}(X)$ for $\mathbb{E}[D|X]$
3. Compute residuals:
 $\tilde{Y}_i = Y_i - \hat{m}_{-k}(X_i)$
 $\tilde{D}_i = D_i - \hat{\ell}_{-k}(X_i)$
4. Estimate: $\hat{\theta} = \frac{\sum \tilde{D}_i \tilde{Y}_i}{\sum \tilde{D}_i^2}$

Module III: Heterogeneous Treatment Effects (Lectures 10–11)

- ▶ **Why heterogeneity matters:**
personalized decisions, targeting, policy design
- ▶ **CATE estimation:**
 - ▶ **DR-Learner:** regress DR pseudo-outcomes on X
 - ▶ **R-Learner:** minimize residual-on-residual loss (for continuous treatments)
- ▶ **Model selection:** DR-Loss as a universal scoring rule for comparing CATE models
- ▶ **Interpretation:** distillation trees, SHAP values for black-box CATE models
- ▶ **Validation:** BLP test, GATEs, AUTOC, Qini curves, calibration

DR-Learner

1. Estimate nuisance functions $\hat{\mu}_d(x)$, $\hat{e}(x)$ via cross-fitting
2. Construct DR pseudo-outcomes $\hat{\psi}_i$
3. Regress $\hat{\psi}_i$ on X_i using any ML method to estimate $\hat{\tau}(x)$

Key Insight

DR-Loss enables model selection for CATE *without* access to ground-truth treatment effects.

Module IV: Causal Graphs & Structure (Lectures 12–13)

- ▶ **SEMs:** non-parametric structural equation models encode causal mechanisms
- ▶ **DAGs:** represent causal structure visually
- ▶ **SWIGs:** formalize interventions; connect to potential outcomes
- ▶ **d-Separation:** graphical criterion for conditional independence (chains, forks, colliders)
- ▶ **Backdoor criterion:** identify valid adjustment sets
- ▶ **Good vs. bad controls:** principled variable selection

Fork (open)



Chain (open)



Collider (blocked)



Module V: Dealing with Unobserved Confounding (Lectures 14–16)

- ▶ **Sensitivity analysis** (L14):
 - ▶ Omitted variable bias formula
 - ▶ Robustness value: how strong must unobserved confounding be to nullify the result?
- ▶ **Front-door criterion** (L14):
identification through mediators even with unobserved confounders
- ▶ **Instrumental variables** (L15):
 - ▶ LATE for heterogeneous effects
 - ▶ DR-LATE, DML-IV
 - ▶ Weak instrument diagnostics
- ▶ **Proximal causal inference** (L16): use noisy proxies of unobserved confounders; DML-Proximal algorithm

Identification Toolkit

Method	Key Assumption
Conditioning	Ignorability + Overlap
Sensitivity	Bounded confounding
Front-door	Mediator structure
IV	Exclusion restriction
Proximal	Proxy availability

Module VI: Unobserved Confounding Beyond DAGs (Lecture 17)

When DAG-based strategies are insufficient:

- ▶ **Difference-in-Differences (L17):**
 - ▶ Parallel trends assumption
 - ▶ Doubly robust DiD estimator
 - ▶ Staggered adoption designs
 - ▶ Combines panel structure with DR methods
- ▶ **Regression Discontinuity (L17):**
 - ▶ Sharp RDD: treatment deterministic at cutoff
 - ▶ Fuzzy RDD: treatment probability jumps at cutoff (IV interpretation)
 - ▶ Local polynomial regression near the threshold
 - ▶ Smoothness assumption at cutoff

DiD: Core Idea

$$\tau_{DiD} = \underbrace{(\bar{Y}_{1,post} - \bar{Y}_{1,pre})}_{\text{Treated change}} - \underbrace{(\bar{Y}_{0,post} - \bar{Y}_{0,pre})}_{\text{Control change}}$$

Removes time-invariant unobserved confounders under parallel trends.

RDD: Core Idea

$$\tau_{RDD} = \lim_{x \downarrow c} \mathbb{E}[Y|X = x] - \lim_{x \uparrow c} \mathbb{E}[Y|X = x]$$

Identifies a local treatment effect at the cutoff c by comparing units just above and below the threshold.

Module VII: Time-Varying Considerations (Lectures 18–19)

- ▶ **Dynamic treatment effects (L18):**
 - ▶ Time-varying confounding: past outcomes affect future treatment
 - ▶ Sequential ignorability assumption
 - ▶ Sequential g-formula via backwards recursion
 - ▶ Dynamic DR/DML: ML-powered estimation of dynamic causal effects
- ▶ **Surrogates for long-term effects (L19):**
 - ▶ Surrogacy assumption: $Y \perp\!\!\!\perp D \mid S, X$
 - ▶ Surrogate index: predict long-term outcome from short-term proxies
 - ▶ DR surrogate estimator: combines surrogate prediction with DR correction
 - ▶ Enables evaluation of long-term effects from short-term experiments

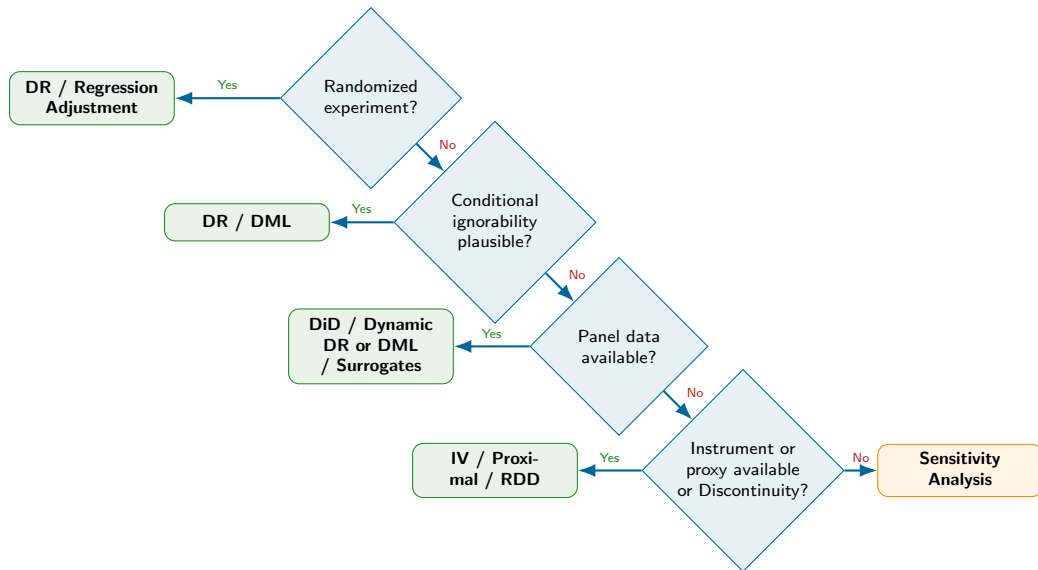
Dynamic DR: Core Idea

Use **backwards recursion** to sequentially estimate effects at each time step and adjust outcomes.
Dynamic DR / DML allows for ML based nuisance estimation.

Surrogates: Key Idea

Use short-term outcomes S to predict long-term outcome Y , then apply DR estimation on the predicted long-term outcome.

A Rough Applied Causal Inference Decision Tree



Key Takeaways

What to Remember from MS&E 228

1. **Causal inference requires assumptions**, not just data. Always state and defend your identifying assumptions.
2. **Doubly robust methods** provide insurance against misspecification—the workhorse of modern causal ML.
3. **Cross-fitting** is essential when using flexible ML for nuisance estimation—prevents overfitting bias.
4. **Heterogeneity is actionable**: CATE enables personalized treatment rules; validate with BLP, GATEs, Qini.
5. **Graphical models** (DAGs/SWIGs) help reason about which variables to condition on and which to avoid. A visual tool to translate interpretable domain knowledge into identification strategies.
6. **When ignorability fails**, you have options: sensitivity analysis, IV, proxies, DiD, RDD.
7. **ML is a tool, not a substitute for causal reasoning**. Domain knowledge guides identification; ML improves estimation.

Quick End of Course Quiz

Quiz Question 1: Potential Outcomes & Identification

Question 1

Consider a binary treatment $D \in \{0, 1\}$ and potential outcomes $Y(0), Y(1)$. Which of the following is the **conditional ignorability** assumption?

- (A) $Y(0), Y(1) \perp\!\!\!\perp D$
- (B) $Y(0), Y(1) \perp\!\!\!\perp D \mid X$
- (C) $Y \perp\!\!\!\perp D \mid X$
- (D) $\mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0]$

(Lectures 1–2)

Quiz Question 2: Bad Controls

Question 2

Suppose $D \rightarrow M \rightarrow Y$ and $D \rightarrow Y$, where M is a mediator. You condition on M when estimating the effect of D on Y . What happens?

- (A) You recover the total causal effect of D on Y
- (B) You block the indirect path and only estimate the direct effect $D \rightarrow Y$
- (C) You introduce collider bias
- (D) You violate SUTVA

(Lectures 2, 13)

Quiz Question 3: Doubly Robust Estimation

Question 3

The doubly robust (DR) estimator for the ATE is consistent if:

- (A) Both the outcome model $\hat{\mu}_d(x)$ and the propensity score $\hat{e}(x)$ are correctly specified
- (B) Either the outcome model or the propensity score is correctly specified (but not necessarily both)
- (C) The propensity score is correctly specified, regardless of the outcome model
- (D) The outcome model is correctly specified, regardless of the propensity score

(Lecture 4)

Quiz Question 4: Cross-Fitting

Question 4

Why is cross-fitting essential when using flexible ML models (e.g., random forests, neural nets) for nuisance estimation in causal inference?

- (A) It increases the sample size available for estimation
- (B) It ensures the nuisance estimates are independent of the observations used to form the causal estimator, preventing overfitting bias
- (C) It guarantees that the ML model achieves the minimax-optimal prediction rate
- (D) It is only needed when the propensity score is unknown

(Lectures 4–5)

Quiz Question 5: FWL Theorem & DML

Question 5

The Frisch-Waugh-Lovell (FWL) theorem states that in the regression $Y = \theta D + X'\beta + \varepsilon$, the coefficient θ can equivalently be obtained by:

- (A) Regressing Y on D without any controls
- (B) Regressing the residual of Y on X against the residual of D on X
- (C) Regressing Y on the fitted values of D from a regression on X
- (D) Regressing D on Y and X simultaneously

(Lecture 8)

Quiz Question 6: CATE Estimation

Question 6

In the DR-Learner approach for estimating the Conditional Average Treatment Effect (CATE), the second stage involves:

- (A) Fitting a propensity score model and reweighting
- (B) Regressing the doubly robust pseudo-outcomes $\hat{\psi}_i$ on covariates X_i using any ML method
- (C) Comparing outcomes between matched treated and control units
- (D) Running a standard OLS regression of Y on D and X

(Lecture 10)

Quiz Question 7: d-Separation

Question 7

Consider the DAG: $D \rightarrow Z \leftarrow Y$. The variable Z is a **collider** on the path $D \rightarrow Z \leftarrow Y$. What happens if you condition on Z ?

- (A) The path $D \rightarrow Z \leftarrow Y$ becomes blocked
- (B) The path $D \rightarrow Z \leftarrow Y$ becomes open, creating a spurious association between D and Y
- (C) Nothing changes because colliders are always open
- (D) Z becomes an instrument for D

(Lecture 13)

Quiz Question 8: Instrumental Variables

Question 8

An instrumental variable Z for the effect of D on Y must satisfy three conditions. Which of the following is **NOT** one of them?

- (A) Z is associated with D (relevance)
- (B) Z affects Y only through D (exclusion restriction)
- (C) Z is independent of unobserved confounders (exogeneity)
- (D) Z must be a continuous variable

(Lecture 15)

Quiz Question 9: Difference-in-Differences

Question 9

The key identifying assumption in Difference-in-Differences (DiD) is:

- (A) The treatment and control groups have the same average outcomes in the pre-treatment period
- (B) In the absence of treatment, the treated and control groups would have followed the same trend over time (parallel trends)
- (C) The treatment assignment is random conditional on observed covariates
- (D) There are no time-varying confounders

(Lecture 17)

Quiz Question 10: Dynamic Treatment Effects

Question 10

In dynamic treatment regimes, why can we **not** simply condition on all time-varying covariates to estimate the effect of a sequence of treatments?

- (A) Because time-varying covariates are always unobserved
- (B) Because a time-varying covariate can be both a confounder for future treatment and a mediator of past treatment, so conditioning on it blocks causal pathways
- (C) Because dynamic settings violate SUTVA
- (D) Because the propensity score cannot be estimated in panel data

(Lecture 18)

Ask Me Anything Q&A

Question 1

DiD Packages, Propensity Score Optimizers & Model Selection

Q1: DiD Packages Produce Different Estimates

Student Question

Scott Cunningham just posted something this morning showing how DiD packages for the Callaway-Sant'Anna estimator (R, Python, Stata) produce different estimates for the ATT when you have covariate sets and common support fails. It seems like the underlying issue is related to how the packages choose to model the propensity score, and what the optimizers do for near separation. Given that we're making choices on our optimizers when implementing DR or Doubly Robust / Double ML, should we have the same concerns about our (chosen) optimizer assumptions? Is one way to interpret his experiment is that these packages would all benefit from an under-the-hood selection of the propensity model via CV / Flaml?

Q1: Response — Optimizer Sensitivity in Propensity-Based Methods

- ▶ Near-separation in the propensity score (some $\hat{e}(x) \approx 0$ or 1) leads to **extreme weights** in IPW and DR estimators
- ▶ This is **not unique to DiD**—it applies to any DR, IPW, or DML estimator that relies on propensity scores

Practical recommendations:

- ▶ Use **cross-validated model selection** (e.g., FLAML) for propensity models
- ▶ Plot **propensity diagnostics**
- ▶ **Trim or clip** extreme propensity scores

Takeaway

Doubly robust estimators are more resilient to propensity score misspecification than pure IPW. Automated model selection for nuisance functions is good practice. Propensity diagnostics and trimming is important with extreme propensities.

References

- ▶ Cunningham (2025), “Apple-to-Apple Audit of Six CS Packages,” *Substack*
- ▶ Callaway & Sant’Anna (2021), *J. Econometrics*
- ▶ Baker et al. (2025), “Difference-in-Differences Designs: A Practitioner’s Guide”

Question 2

Convergence Rate Conditions & Constructing Causal DAGs

Q2: Interpreting Convergence Rates & Building DAGs

Student Question

I'm curious how interpretable convergence rate conditions (e.g. product rate condition, propensity converging at a rate of $O(n^{-1/4})$, and so on) are in practice. How do we test and interpret these conditions, particularly the more complex ones? Can we use these convergence rates to compare methods?

On a similar note, I'm also curious on how to best create causal DAGs in practice. The estimated effect can differ depending on what is included in the DAG and how it is constructed, and even if we consult domain experts, two domain experts might say different things. How can we go about creating DAGs in a structured, rigorous manner to ensure our estimated effects are sensible? Thank you!

Q2a: Response — Convergence Rates in Practice

- ▶ The **product rate condition** (e.g., $\|\hat{\mu} - \mu\| \cdot \|\hat{e} - e\| = o(n^{-1/2})$) is a *sufficient condition* for $\sqrt{n}$ -consistency of DR/DML estimators
- ▶ In practice, we **cannot directly test** these rates from finite data
- ▶ Instead, we rely on:
 - ▶ Known rates for specific ML methods (e.g., forests achieve $n^{-1/4}$ under smoothness)
 - ▶ **Cross-validation** performance as a proxy for prediction quality
 - ▶ **Sensitivity checks**: vary the ML method and check stability of the causal estimate
- ▶ The product rate is *weaker* than requiring each model to converge at $n^{-1/4}$ —doubly robust methods exploit this

References

One notable attempt at "testing" and "rejecting" the product rate:

- ▶ Liu, L., Mukherjee, R., & Robins, J. M. (2024). Assumption-lean falsification tests of rate double-robustness of double-machine-learning estimators. *Journal of econometrics*, 240(2), 105500.

Q2b: Response — Constructing Causal DAGs in Practice

- ▶ **Sensitivity to DAG specification:** if two plausible DAGs imply different adjustment sets, report results under both
- ▶ **Causal discovery algorithms** (PC, GES, NOTEARS) can suggest structure from data, but require strong assumptions (faithfulness, no latent variables)
- ▶ **LLM based causal discovery** LLMs can bring domain knowledge and propose DAGs that many times are at the level of what an expert could produce.

Best Practice

Use DAGs as a *transparency device*: make your assumptions explicit and testable. When experts disagree, treat the DAG as uncertain and conduct sensitivity analysis over alternative structures. Augment classical methods with LLM / Agentic elicitation.

References

- ▶ Rodrigues et al. (2022), "Constructing DAGs with Domain Experts," *IJE*
- ▶ Sawarni, A., Tan, J., & Syrgkanis, V. (2026). CausalReasoningBenchmark. arXiv preprint arXiv:2602.20571.
- ▶ Jin, Z., Chen, Y., Leeb, F., Gresele, L., Kamal, O., Lyu, Z., ... & Schölkopf, B. (2023). Cladder: Assessing causal reasoning in language models. NeurIPS

Question 3

Control Selection for ATE vs. CATE with DML

Q3: Which Controls to Include in ATE/CATE Analysis?

Student Question

It might sound like a silly question considering that we had a few classes about selecting good controls, but in practice, starting from this subset of good controls, which of them should we include when running ATE/ATT/CATE analysis using DML or other ML methods?

For example, in the past I worked on an RCT where we were trying to estimate treatment effect heterogeneity using GenericML (method designed for Prof. Chernozhukov), and we often had the question of whether it is fine to include all good controls, or if it is preferable to include only a subgroup of them that might explain something in the data (for example, in my case a subset of psychological variables that might explain entrepreneurial success, rather than everything).

Is it generally fine to include all good controls? Does the answer change if we are interested in estimating the ATE or CATE?

Q3: Response — Variable Selection for ATE vs. CATE

- ▶ **Include all good controls** (pre-treatment variables that satisfy the backdoor criterion)
- ▶ ML-based nuisance estimation (forests, boosting) performs implicit variable selection—irrelevant controls are down-weighted
- ▶ The **CATE model** (second stage) may benefit from a more targeted set of *effect modifiers*—variables that plausibly interact with treatment
- ▶ DR-Loss and cross-validation can guide this selection

Key Distinction

Nuisance estimation (stage 1): use all good controls for best prediction.

CATE estimation (stage 2): focus on variables that are plausible effect modifiers. ML handles the selection, but domain knowledge helps.

CATE validation (stage 3): Always run validation checks for your CATE model.

Question 4

Power for Detecting Treatment Effect Heterogeneity

Q4: Do You Need More Power for CATE Than ATE?

Student Question

Do you need more power to detect heterogeneous treatment effects than average treatment effects? Following the previous example, I remember that Prof. Chernozhukov mentioned in a seminar that you need around 5,000 observations to detect heterogeneity using GenericML with neural networks. However, I have also seen papers detecting heterogeneity with around 1,000 observations using his method (which relies on sample splitting, where treatment effects are estimated using half of the sample).

These sample sizes might be feasible in tech companies, but in the social sciences each observation is expensive and reaching large samples is often difficult. So, following this question, do we generally need more observations to detect heterogeneity than to estimate average treatment effects? What would be an ideal sample size in practice?

Q4: Response — Power Requirements for Heterogeneity Detection

Yes, detecting heterogeneity requires substantially more power than estimating the ATE:

- ▶ **ATE** is a single scalar—you average over all units, so variance decreases at rate $1/n$
- ▶ **CATE** is a function of X —estimating it well requires learning a *conditional* relationship, which is harder
- ▶ **Sample splitting** (required for valid inference) further reduces effective sample size
- ▶ The $\sim 5,000$ figure is for *detecting* heterogeneity with complex ML models (neural nets); simpler models (linear CATE) need less

Rules of thumb

- ▶ $n \geq 1,000$: may detect strong, low-dimensional heterogeneity
- ▶ $n \geq 5,000$: reasonable for flexible ML-based CATE
- ▶ Signal strength matters as much as n

Question 5

Untapped Potential for Causal Inference

Q5: Where Can Causal Inference Have Greater Impact?

Student Question

In what fields (e.g., policy, healthcare, economics) do you think there is a lot of untapped potential for causal inference to have a greater impact?

Q5: Response — High-Impact Frontiers for Causal ML

Fields with significant untapped potential:

- ▶ **Climate & sustainability:** causal effects of environmental regulations; evaluating green technology adoption
- ▶ **Technology & AI:** causal reasoning in LLMs; causal inference for LLM / Agentic performance evaluation
- ▶ **Healthcare & precision medicine:** personalized treatment rules using CATE; dynamic treatment regimes for chronic diseases; surrogate-based evaluation of long-term drug effects
- ▶ **Public policy:** heterogeneous effects of education, welfare, and criminal justice interventions; optimal targeting of limited resources

Question 6

Analysis When Key Confounders Are Unobserved

Q6: How to Handle Unobserved Confounders?

Student Question

In many real-world settings, the assumption of unconfoundedness seems difficult to satisfy. How do we do analysis when key confounders are unobserved?

Q6: Response — A Toolkit for Unobserved Confounding

We covered several strategies in this course:

- ▶ **Sensitivity analysis** (L14): quantify how strong unobserved confounding must be to overturn your result; report robustness values
- ▶ **Instrumental variables** (L15): exploit an exogenous source of variation; identifies LATE for compliers
- ▶ **Proximal causal inference** (L16): use noisy proxies of the unobserved confounder; requires two proxies (treatment-side and outcome-side)
- ▶ **Difference-in-Differences** (L17): exploit panel structure; removes time-invariant confounders under parallel trends
- ▶ **Regression discontinuity** (L17): exploit sharp treatment thresholds for local identification
- ▶ **Front-door criterion** (L14): identify effects through mediators when confounders are unobserved

Question 7

The Future of Research in Causal ML

Q7: What's Next for Research in Causal ML?

Student Question

What's next for research in causal ML? What are the gaps that the newest research has been filling?

Q7: Response — Frontiers and Open Problems in Causal ML

- ▶ **Causal inference + LLMs:** using language models for causal reasoning, DAG construction, and as proxies for unobserved confounders (text as treatment/confounder)
- ▶ **Causal discovery at scale:** learning causal structure from high-dimensional observational data; combining domain knowledge with data-driven methods
- ▶ **Robust and adaptive experimentation:** bandits with causal constraints; adaptive designs that balance exploration and exploitation
- ▶ **Long-term and dynamic effects:** surrogates, dynamic regimes, and off-policy evaluation in sequential settings
- ▶ **Causal representation learning:** learning causal variables from raw data (images, text)

Thank You!

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Keep exploring:

<https://causalml-book.org> — Course textbook

[EconML](https://pywhy.org/EconML) (pywhy.org/EconML) — Python library for causal ML

[DoubleML](https://docs.doubleml.org) (docs.doubleml.org) — DML implementation

Best of luck in your future research and careers!